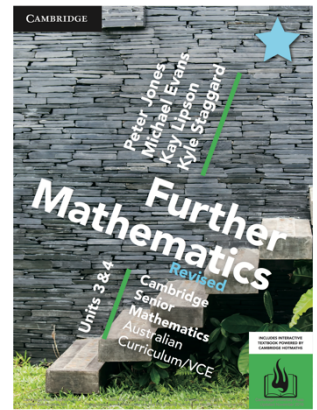




Further Mathematics 2018

CORE: Data analysis

Chapter 2 – Summarising numerical data



Extract from Study Design

Key knowledge

- Types of data: categorical (nominal and ordinal) and numerical (discrete and continuous)
- Frequency tables, bar charts including segmented bar charts, histograms, stem plots, dot plots, and their application in the context of displaying and describing distributions
- log (base 10) scales, and their purpose and application
- Five-number summary and boxplots (including the designation and display of possible outliers)
- Mean $\bar{x}$ and standard deviation s_x
- Normal model and the 68–95–99.7% rule, and standardised values (z-scores)

Key skills

- Construct frequency tables and bar charts and use them to describe and interpret the distributions of categorical variables
- Answer statistical questions that require a knowledge of the distribution/s of one or more categorical variables
- Construct stem and dot plots, boxplots, histograms and appropriate summary statistics and use them to describe and interpret the distributions of numerical variables
- Answer statistical questions that require a knowledge of the distribution/s of one or more numerical variables
- Solve problems using the z-scores and the 68–95–99.7% rule

Chapter Sections	Questions to be completed
2A Dot plots and stem plots	1, 2, 3, 4
2B The median, range and interquartile range (IQR)	1, 2, 3, 4, 5
2C The five-number summary and the box plot	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
2D Relating a box plot to shape	1
2E Using box plots to describe and compare distributions	1, 2, 3
2F Describing the centre and spread of symmetric distributions	1, 2, 3, 4, 5, 6, 7, 8 1, 2, 3, 4, 5, 6
2G The normal distribution and the 68-95-99.7% rule	1, 2, 3, 4, 5
2H Standard scores	1, 2, 3, 4
2I Populations and Samples	
Chapter 2 Review	All questions

MORE RESOURCES

<http://drweiser.weebly.com>



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2A Dot plots and stem plots

The dot plot

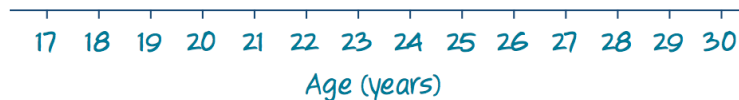
- Suitable for displaying discrete data.

Example 1

The ages (in years) of the 13 members of a cricket team are

22 19 18 19 23 25 22 29 18 22 23 24 22

Construct a dot plot.



The stem plot

- Stem plot works for discrete and continuous data.
- Stem plot is made of two parts: its leading digits, which are the stem, and its last digit which are the leaf.
- Last digit is always the leaf eg. 501, 512, 511
- Must** be **in order** from **lowest** to **highest**
- Must have a key**
- If bunched, break stem into halves or fifths (i.e. if Leaf is too long)

stem	leaf
50	1
51	1 2

Example

Plot the following data in a stem and leaf plot of: (a) halves, (b) fifths.

50 51 53 53 54 55 55 56 56 57 59

(a) Halves

Stem	Leaf	Range	
5	0 1 3 3 4	← (50-54) lower half	Key: 5 0 = 50
5*	5 5 6 6 7 9	← (55-59) upper half	

(b) Fifths

Stem	Leaf	Range	
5	0 1	← (50-51)	Key: 5 9 = 59
5	3 3	← (52-53)	
5	4 5 5	← (54-55)	
5	6 6 7	← (56-57)	
5	9	← (58-59)	

Example 2

University participation rates (%) in 23 countries are given below.

26 3 12 36 1 25 26 13 9 26 27 15 21 7 8 22 3 37 17 55 30 1

Display the data in the form of a stem plot.

Stem	Leaf
0	
1	
2	
3	
4	
5	

Key: | = _____

2B Median, range and interquartile range (IQR)

The most useful tools for numerically describing the centre and spread of a distribution are:

- the **median** (the middle value)
- the **range** (the maximum spread of the data)
- the **interquartile range** (the spread of the middle half of the data)

Determining the median

To find the median in an order set of values

median is located at the $\left(\frac{n+1}{2}\right)$ th position

- When n is **odd**, the median will be the middle data value
- When n is **even**, the median will be the average of the two middle data values.

Example 3

Order each of the following datasets, locate the median, and then write down its value.

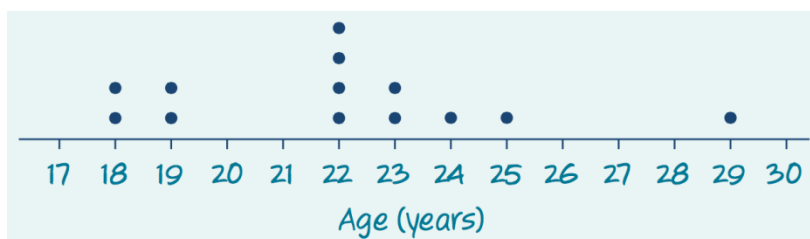
a) 2, 9, 1, 8, 3, 5, 3, 8, 1

b) 10, 1, 3, 4, 8, 6, 10, 1, 2, 9

Using a dot plot to help locate medians

Example 4

The dot plot displays the age distribution (in years) of the 13 members of a local cricket team.



Determine the median age of these cricket and mark its location on the dot plot.

Example 5

The stem plot opposite displays the maximum temperature (in °C) for 12 days in January. Determine the median maximum temperature for these 12 days.

Key: 0|8 = 8°C

1	8 9 9
2	0 2 5 7 8 9 9
3	1 3

The range

The range, R , is the simplest measure of spread of a distribution. It is the difference between the largest and smallest values in the dataset.

$$R = \text{largest data value} - \text{smallest data value}$$

Example 6

The stem plot (in example 5) displays the maximum temperature (in °C) for 12 days in January. Determine the temperature range over these 12 days.

The interquartile range

- More useful to determine spread than the range, as it is not influenced by outliers and doesn't account for how data is spread out in between the minimum and maximum values.
- IQR is the spread of the middle 50% of data values.

$$IQR = Q_3 - Q_1$$

Example 7

Use the stem plot to determine the quartiles Q_1 and Q_3 , the IQR and the range, R , for life expectancies. The median life expectancy is $M=73$.

Stem: 5|2 = 52 years

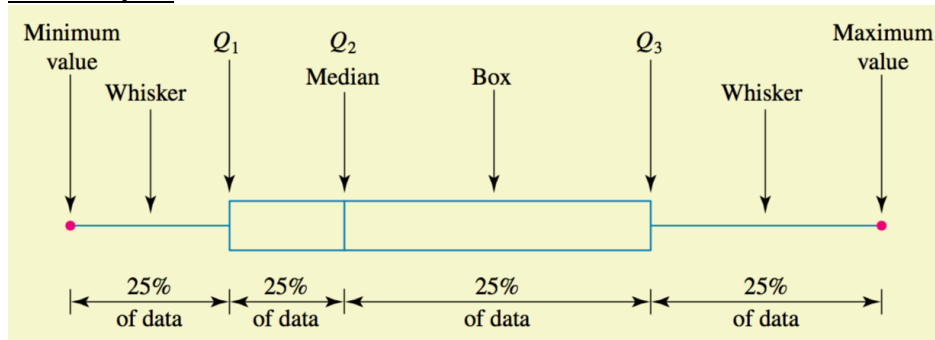
5	2
5	5 6
6	4
6	6 6 7 9
7	1 2 2 3 3 4 4 4 4
7	5 5 6 6 7 7

2C The five-number summary and the box plot

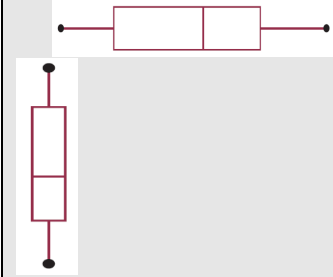
The five-number summary

- minimum, Q_1 , median, Q_3 , maximum.

The box plot



Boxplots can be drawn horizontally or vertically.



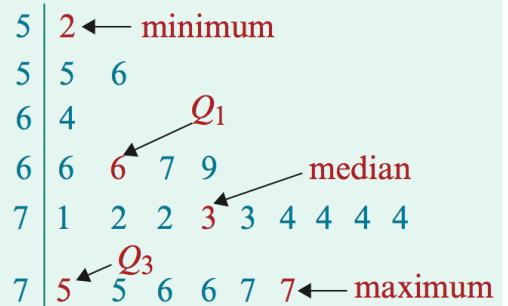
Example 8

The stem plot shows the distribution of life expectancies (in years) in 23 countries.

The five-number summary for the data is:

minimum	52
first quartile (Q_1)	66
median (M)	73
third quartile (Q_3)	75
maximum	77

Key: 5|2 = 52 years



Use the five-number summary to construct a box plot.



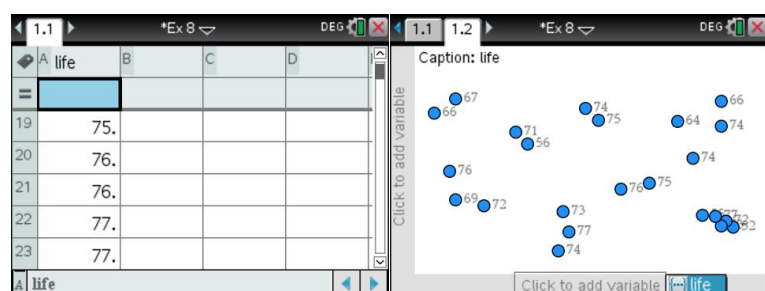
Example 8 (CAS Calculator)

On a List & Spreadsheet page, Enter the data values in column A.

Then press

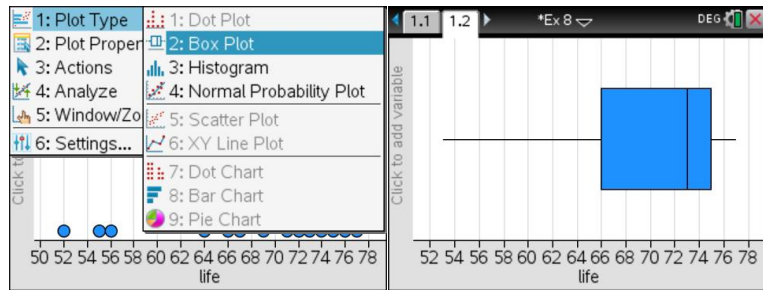
- **ctrl** **doc** and add a data & statistics page

Click to add variable and choose the labelled column "life"



Now press

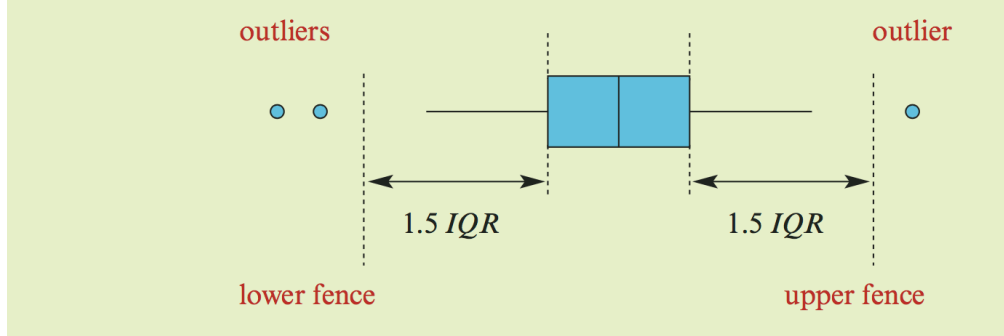
- **menu** **1** **2** for a boxplot



Box plots with outliers

To display outliers on box plots we need to determine the upper and lower fence.

- greater than $Q_3 + 1.5 \times IQR$ (upper fence)
- less than $Q_1 - 1.5 \times IQR$ (lower fence).



CAS Example Box plot with Outliers

Display the following set of 19 marks in the form of a box plot with outliers.

28 21 21 3 22 31 35 26 27 33 43 31 30 34 48 36 35 23 24

On a List & Spreadsheet page, Enter the data values in column A.

Then press

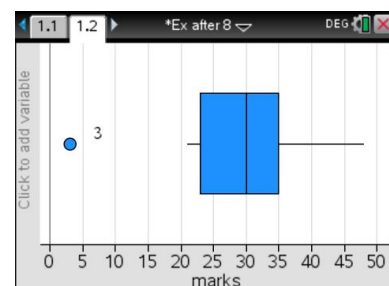
- **ctrl** **doc** and add a data & statistics page

	A	B	C	D
1	28.			
2	21.			
3	21.			
4	3.			
5	22.			

Click to add variable and choose the labelled column "marks"

Now press

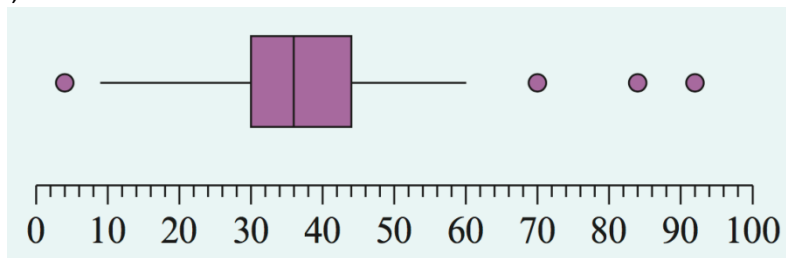
- **menu** **1** **2** for a boxplot



Note: The CAS calculator works out any outliers internally and displays them.

Interpreting box plots**Example 9**

For the box plot shown, write down the values of:



a) The median

b) Q_1 and Q_3

c) The IQR

d) The minimum and maximum values

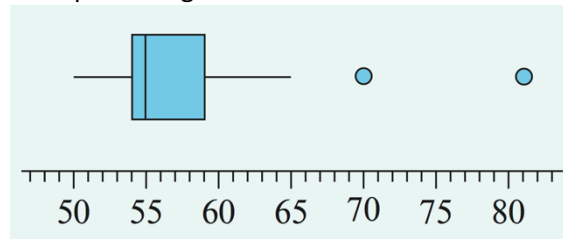
e) The values of any possible outliers

f) The smallest value in the upper end of the dataset that will be classified as an outlier.

g) The largest value in the lower end of the dataset that will be classified an outlier.

Example 10

For the box plot shown, estimate the percentage of values:



a) Less than 54

b) Less than 55

c) Less than 59

d) Greater than 59

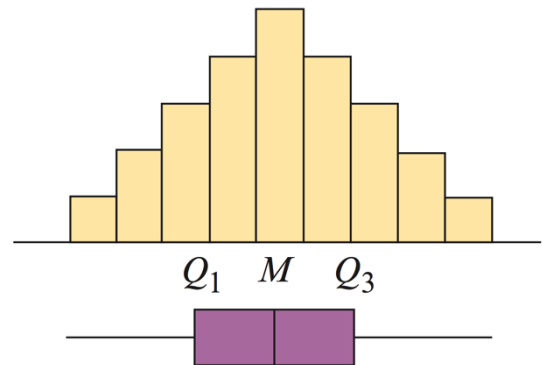
e) Between 54 and 59

f) Between 54 and 86

2D Relating a box plot to shape

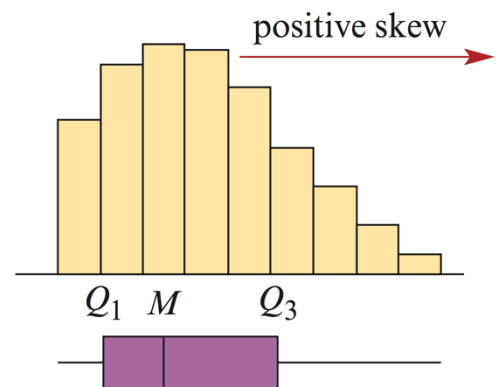
A symmetric distribution

- Centred on its median
- Values evenly spread around median
- Box plot will be symmetric
- Median close to middle of box and whiskers will be approximately equal
- Mean will be approximately the same as the median



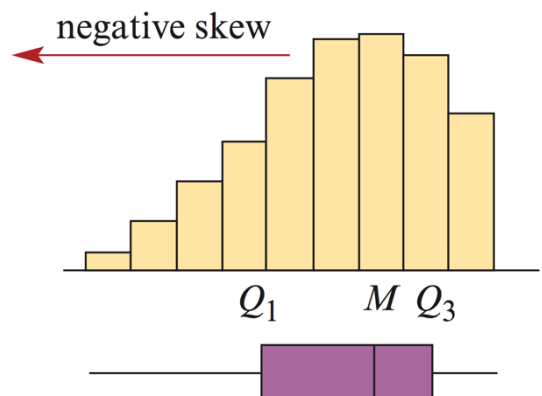
Positively skewed distributions

- Cluster of values around median on left-hand side of distribution
- Tail off to the right
- Box plot will have the median off to the left-hand side of the box
- Left-hand whisker will be shorter with the right-hand whisker longer
- Median is used to measure the centre of the data rather than mean



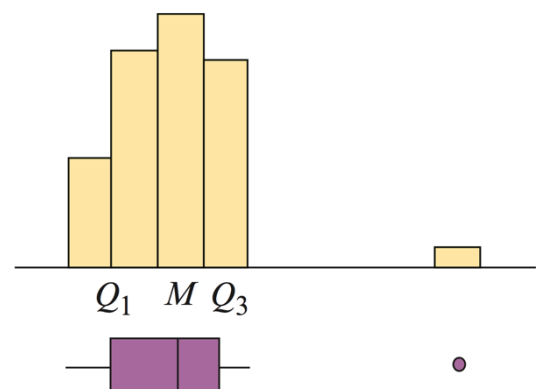
Negatively skewed distributions

- Cluster of values around the median on the right-hand side of the distribution
- Tail off to the left
- Box plot will have the median off to the right-hand side of the box
- Right-hand whisker will be shorter with the left-hand whisker longer
- Median is used to measure the centre of the data rather than mean



Distributions with outliers.

- Characterised by large gaps between main body and data values in the tails.
- The outlier is represented by a dot that is separate from the box and whiskers

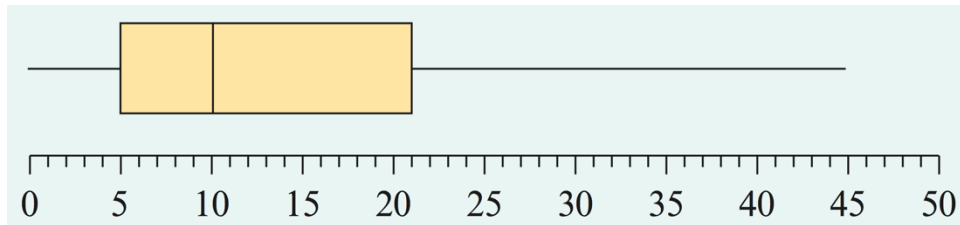


2E Using box plots to describe and compare distributions

The information contained in a boxplot make it a powerful tool for describing a distribution in terms of shape, centre and spread.

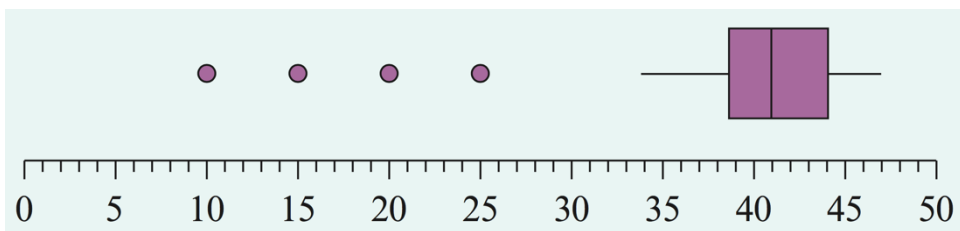
Example 11

Describe the distribution represented by the box plot in terms of shape, centre and spread. Give appropriate values.



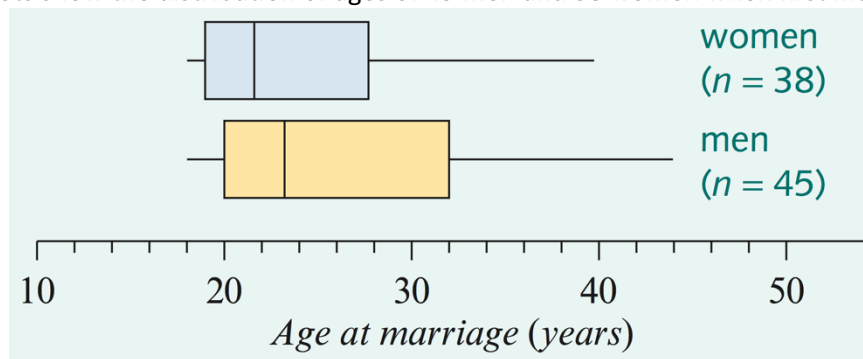
Example 12

Describe the distributions represented by the box plot in terms of shape and outliers, centre and spread. Give appropriate values.



Example 13

The parallel box plots show the distribution of ages of 45 men and 38 women when first married.



- a) Compare the two distributions in terms of shape (including outliers, if any), centre and spread. Give appropriate values at a level of accuracy that can be read from the plot.

- b) Comment on how the age of the men when first married compares to that of women.

2F Describing the centre and spread of symmetric distributions

The mean

The mean of a set of data is what most people call the 'average'. The mean of a set of data is given by:

$$\text{mean} = \frac{\text{sum of data values}}{\text{total number of data values}}$$

or

$$\bar{x} = \frac{\sum x}{n}$$

where $\bar{x}$ is pronounced 'x bar' and the Greek Σ means 'sum of'

Example 14

The following is a set of reaction times (in milliseconds): 38, 36, 35, 43, 46, 64, 48, 25

a)	n	b)	b) Σx	c)	c) $\bar{x}$

When to use the median rather than the mean

Because the value of the *median* is relatively unaffected by the presence of extreme values in a distribution, it is said to be a *resistant* statistic. For this reason, the median is frequently used as a measure of centre when the distribution is known to be clearly *skewed* and/or likely to contain *outliers*.

The standard deviation

- To measure the spread of data around the median we use the IQR.
- To measure the spread of data around the mean we use standard deviation.

The formula for standard deviation, s , is:

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$

How to calculator the mean and standard deviation using the CAS calculator

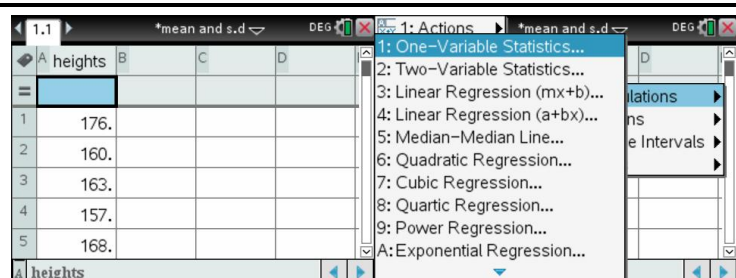
The following are the heights (in cm) of a group of women.

176 160 163 157 168 172 173 169

Determine the mean and standard deviation of the women's heights. Give your answers correct to two decimal places.

On a Lists & Spreadsheet page, enter the data into column A and label it 'height'.

- Highlight the column and then press:
- Menu  > **4** Statistics > **1** Stat Calculations > **1** One-Variable Statistics

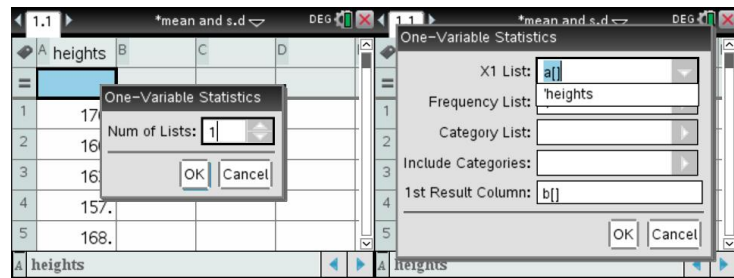


Press **tab** to ok and press **enter**

Enter 'heights' into the X1 list

tab to the ok press **enter** to generate the summary statistics.

Scroll down to look for $\bar{x} = 19.3$



Scroll down to look for:

- the mean $\bar{x} = 167.25$
- and $s_x = 6.67\text{cm}$

1.1	*mean and s.d			DEG
A	heights	B	C	D
=			=OneVar(	
1	176.	Title	One-Va...	
2	160.	$\bar{x}$	167.25	
3	163.	Σx	1338.	
4	157.	Σx^2	224092.	
5	168.	$s_x := s_n \dots$	6.67083...	
CI = "One-Variable Statistics"				

2G The normal distribution and the 68-95-99.7% rule

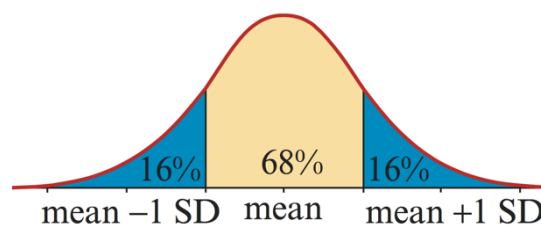
The normal distribution

Many datasets are roughly symmetrical and have an approximately bell shaped curve. Data distributions that are bell shaped can be modelled by a normal distribution.

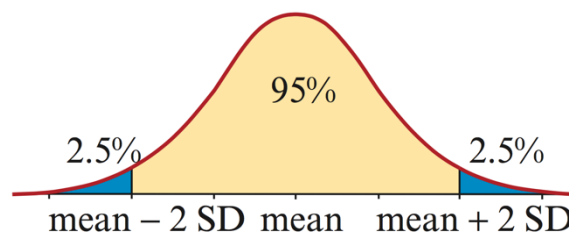
The 68-95-99.7% rule

For a normal distribution, approximately:

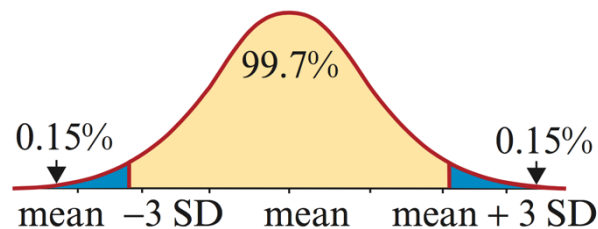
- 68% of the observations lie within *one* standard deviation of the mean



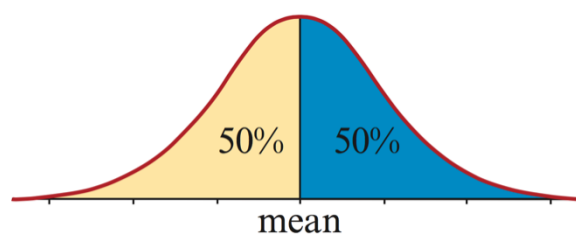
- 95% of the observations lie within *two* standard deviations of the mean



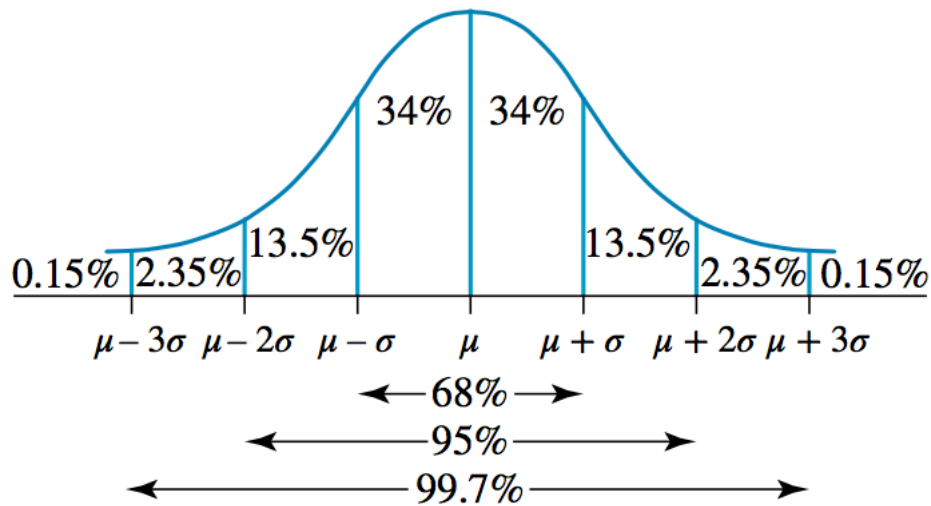
- 99.7% of the observations lie within *three* standard deviations of the mean



- 50% of the data values will lie *above* the mean and 50% of values will lie *below* the mean.



Combining all this information gives the following:



Example 15

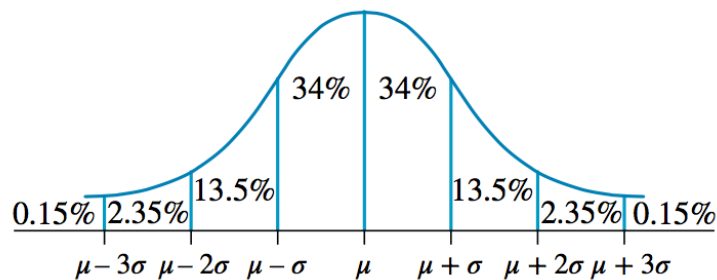
The distribution of delivery times for pizzas made by House of Pizza is approximately normal, with a mean of 25 minutes and a standard deviation of 5 minutes.

a) What percentage of pizzas have delivery times of between 15 and 35 minutes?

i. Identify the mean and standard deviation

- Mean = _____
- St Dev = _____

ii. Label the distribution below



b) What percentages of pizzas have delivery times of greater than 30 minutes?

c) In 1 month, House of Pizza delivers 2000 pizzas. How many of these pizzas are delivered in less than 10 minutes?

2H Standardized scores

z-score

The **z-score** (also called the **standardised score**) is used to measure the position of a score in a data set relative to the mean.

To obtain a standard score for an actual score, subtract the mean from the score and then divide the result by the standard deviation. That is:

$$\text{standard score} = \frac{\text{actual score} - \text{mean}}{\text{standard deviation}} \quad \text{or} \quad z = \frac{x - \bar{x}}{s}$$

- a **positive** z-score indicates that **the actual score** it represents **lies above the mean**
- a **zero** standardised score indicates that **the actual score** is **equal to the mean**
- a **negative** z-score indicates that **the actual score** lies **below the mean**.

Example 16

The heights of a group of young women have a mean of 160 cm and a standard deviation of 8 cm. Determine the standard of z-scores of a woman who is:

a) 172 cm tall

b) 150 cm tall

c) 160 cm tall

Using standard scores to compare performance

Standard scores are useful for comparing groups that have different means and/or standard deviation.

Example

Stephanie obtained a mark of 75 in Psychology and a mark of 70 in Statistics. In which subject did she do better? In which subject did she do better?

Subject	Mark	Standard	
		Mean	Deviation
Psychology	75	65	10
Statistics	70	60	5

Example 17

Another student studying the same two subjects obtained a mark of 55 for both Psychology and Statistics. Does this mean that she performed equally well in both subjects? Use standardised marks to help you arrive at your conclusion.

Converting standardised scores into actual scores

By making the actual score the subject of the rule for calculating standard scores, we arrive at:

$$\text{actual score} = \text{mean} + \text{standard score} \times \text{standard deviation} \quad \text{or} \quad x = \bar{x} + z \times s$$

Example 18

A class test (out of 50) has a mean mark of 34 and a standard deviation of 4. Joe's standardised test mark was $z = -1.5$. What was Joe's actual mark?

2I Populations and samples

A group of Year 12 decide to investigate how much money all Year 12s spend on birthday presents. It would take a long time to survey all 200 students. So a smaller group known as a *sample* is taken from the total *population* of Year 12.

The mean of a data set which represents a population is μ .

The mean of a data set which represents a sample is $\bar{x}$.

The standard deviation of a data set which represents a population is σ .

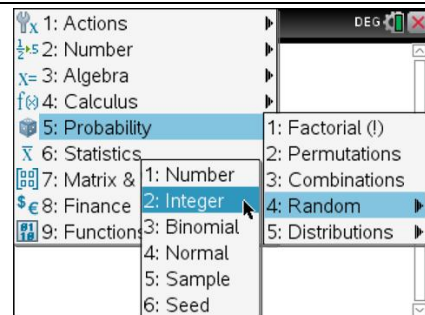
The standard deviation of a data set which represents a sample is s .

Example

Generate 5 random numbers (integers) between 1 and 50.

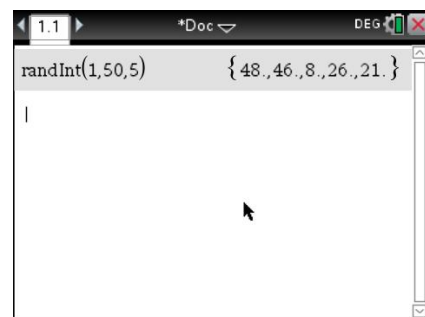
To generate random integers using a CAS calculator, open a Calculator page and press:

- MENU 
- 5: Probability 
- 4: Random 
- 2: Integer 



To generate 5 random numbers between 1 and 50, complete the entry line as: `randInt(1, 50, 5)`.

Then press ENTER .



NORMAL DISTRIBUTION DIAGRAMS FOR USE IN YOUR SACs/EXAMS

This is a support tool for you to use

